# HyperSL: A Spectral Foundation Model for Hyperspectral Image Interpretation

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***Abstract*— This work has delivered a novel foundational model for hyperspectral remote sensing images. Current approaches for hyperspectral data interpretation often require specialized models that are specifically tailored to individual datasets or tasks. In some specific tasks, the availability of hyperspectral data is often limited, posing significant challenges to training due to data insufficiency. Furthermore, the diverse structure of hyperspectral data often complicates the transfer of knowledge from other available datasets, severely limiting cross-scenario capabilities. To bridge this gap, we introduce a highly adaptable foundational model with strong transferability, capable of processing all forms of hyperspectral data across diverse spectral bands and ranges. Compared to previous methods, our approach: 1) standardizes all types spectral vectors into a common token format, enabling a single model for multisource hyperspectral data; 2) aligns spectral features across different ranges by embedding wavelength information into position encoding; 3) has been pretrained on over 300 million spectral instances worldwide, ensuring broad generalization; and 4) transfers learned knowledge effortlessly to downstream tasks and new datasets without architectural modifications or training from scratch. Experimental results demonstrate that, compared to other mainstream methods, our approach achieves state-of-the-art classification performance across various datasets with different spectral characteristics in both supervised and unsupervised learning settings, while also delivering impressive results in change detection tasks. The source code and the pretrained weights are available at https://github.com/kkweil/HyperSL.**

***Index Terms*— Cross-scenario, foundation model, hyperspectral image, knowledge transfer.**

## I. Introduction

IN HYPERSPECTRAL imaging, a spectrometer or filter set decomposes reflected light into multiple spectral bands [1], which are sequentially captured by the imaging sensor to form a multichannel spectral image $X \in R^{W\times H\times C}$, where $W$, $H$, and $C$ represent the width, height, and channel of HSIs. HSIs contain rich spectral information that reveals material properties [2], making them widely applicable in fields such as agriculture [3], environmental monitoring [4], remote sensing, medical diagnostics [5], food safety [6], mineral exploration [7], and Earth observation [8]. However, processing these HSIs often presents several challenges, including: dimensionality disaster, data redundancy, spatial variability [9], diversity of spectral ranges, and inconsistency in data dimensions due to the inherent characteristics of HSIs. The inconsistency in the dimensions and spectral ranges of HSIs is primarily determined by the parameters of the sensors used to capture the hyperspectral data [10]. Additionally, variations in preprocessing methods applied to the data for different application scenarios can lead to differences in data dimensions and spectral ranges, even when the data is captured by the same sensor. Consequently, multisource HSIs exhibit a high degree of inconsistency in spectral dimensions and ranges.

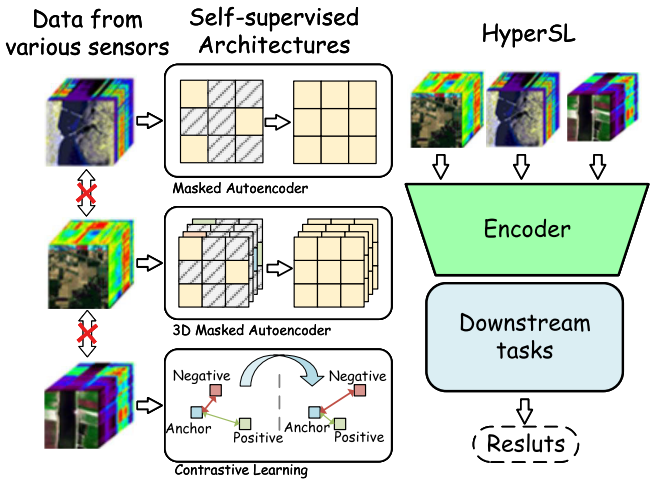



Fig. 1. Most previous foundational models for HSI interpretation are highly scene-specific and tailored to particular scenarios, which limits their generalization across different contexts and sensors. By contrast, our method provides a unified representation mechanism with innate cross-scenario adaptability, enabling the efficient and seamless transfer of pretrained knowledge to downstream tasks.

Representation learning has emerged as a primary approach to address the curse of dimensionality, data redundancy, and spatial variability, driven by the emerging paradigm of compressed sensing [9], [11]. Early representative methodologies included sparse representation [12] and low-rank subspace

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TABLE I
DETAILED CHARACTERISTICS OF THE AVAILABLE PUBLIC DATASETS

| Dataset | Sensor | Band | Spectral Range/**nm** | GSD/**m** | Size | Num |
|---|---|---|---|---|---|---|
| Botswana | Hyperion | 145 | 400-2500 | 30 | 1476×256 | 1 |
| Houston | ITRES CASI-1500 | 144 | 380-1050 | 2.5 | 349×1905 | 1 |
| Indian Pines | AVIRIS | 220 | 400-2500 | 20 | 145×145 | 1 |
| KSC | AVIRIS | 176 | 400-2500 | 18 | 512×614 | 1 |
| NewXiongAn | PHI | 250 | 400-1000 | 0.5 | 3750×1580 | 1 |
| Salinas | AVIRIS | 224 | 400-2500 | 3.7 | 512×217 | 1 |
| Xuzhou | HYSPEX | 436 | 420-2508 | 0.73 | 500×260 | 1 |
| WHU-Hi | Nano-Hyperspec | 274 | 400-1000 | 0.109 | 1217×303, 940×475, 550×400 | 3 |
| Chikusei | Hyperspec-VNIR-C | 128 | 340-1000 | 2.5 | 2517×2335 | 1 |
| WashingtonDC | Hydice | 191 | 400-2400 | - | 1280×307 | 1 |
| Pavia University | ROSIS | 103 | 430-860 | 1.3 | 610×340 | 1 |
| Pavia Center | ROSIS | 102 | 430-860 | 1.3 | 1096×1096 | 1 |
| Cooke City | HyMap | 126 | 450-2480 | 3 | 280×800 | 1 |
| SDFC | HAHS | 63 | 400-1000 | 0.5 | 1201×201 | 1 |
| Hyperview | HYSPEX | 150 | 960-2500 | 2 | avg. 60×60 | 3K |
| HySpecNet-11k | ENMAP | 202 | 420-2450 | 30 | 128×128 | 11K |
| HyperGlobal-450k | EO-1 and GAOFEN-5B | 242 and 330 | 400-2500 | 30 | 64×64 | 447K |
| SpectralEarth | ENMAP | 202 | 400-2500 | 30 | 128×128 | 538K |

representation [13]. With the development of deep learning, many deep learning-based representation methods have achieved remarkable performance in various downstream tasks [14], [15], [16], [17], [18]. For example, Wang et al. [19] proposed Flex-MCFNet with a novel data augmentation strategy for HSI fusion classification. Hu et al. [20] proposed a semi-supervised change detection network that trained jointly with a small amount of labeled data and a large amount of unlabeled data to solve the limitation of training data scarcity. Tian et al. [21] proposed a label disambiguation model for HSI partial label learning, it uses a coherent framework to solve the key problem of the mutual entanglement between representation learning and label disambiguation in HSI partial label learning and achieves the state-of-the-art performance on widely used hyperspectral classification datasets. Among the various deep learning methods, representation methods based on autoencoders have attracted significant interest from researchers due to their remarkable ability to capture nonlinear features [22]. Many excellent works based on autoencoders have emerged in common hyperspectral interpretation tasks, such as [23], [24], and [25]. Although these approaches have proven effective, they are trained on a single dataset, meaning that these methods are limited to a specific dataset. In cross-scenario situations, such as when handling data with different dimensions or data captured by various sensors, these methods become ineffective.

In recent years, pretrained large language models (LLMs) have garnered significant attention across various research fields due to their exceptional generalization capabilities in downstream tasks, such as [26], [27], [28], and [29]. This success is largely attributed to self-supervised pretraining strategies, which effectively utilize massive unannotated training data and large-scale model parameters. In the field of spectral remote sensing image interpretation, preliminary researches on pretrained large-scale foundational models have been initiated in the past two years to enhance generalizability. For example, SpectralGPT [30] utilizes a mask autoencoder (MAE) [31] as a backbone network to create a foundational model for multispectral remote sensing images, meanwhile, SkySense [32] constructs a foundational model trained on multimodal remote sensing data (RGB, MS, SAR, and Temporal). Initial efforts to create large models for HSIs have been made with HyperSIGMA [33] and SpectralEarth [18]. HyperSIGMA uses hyperspectral data that captured from separate sensors for pretraining, employing a random cropping method to address the dimensional inconsistency of HSIs. However, this approach sacrifices some spectral information and significantly alters the data distribution compared to real-world scenarios. SpectralEarth, on the other hand, trains a foundational model using approximately 3.3TB of EnMap hyperspectral data, but its dataset lacks diversity and relies on a simple transformer framework to facilitate adaptability across different sensors. Consequently, to date, there is still no truly general model for HSIs. In this regard, we ask: *What has hindered the progress of this work?* We attempt to analyze this issue from the following perspectives.

1) *Data scarcity:* The number of available HSI datasets is significantly lower compared to other tasks. Previous research had access to only a few dozen HSIs, while datasets in other modalities have reached millions. However, with the increasing deployment of hyperspectral remote sensing satellite missions, the availability and accessibility of HSI are rapidly expanding, with new datasets such as PRISMA, HypSEO, DESIS, Gaofen-5, EnMap, HyspIRI, and so on [34]. Most of this data has been, or will soon be, made publicly available for academic research, thus the data scarcity will no longer be a concern.
2) *Intrinsic multimodal property of HSI:* HSI differs from other remote sensing image modalities in terms of the number of channels, spectral ranges, and resolutions, which vary according to task requirements and imaging mechanisms. This variation means that HSIs collected from different platforms often have varying numbers of

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channels, with each channel corresponding to distinct physical spectral characteristics. The intrinsic, non-fixed multimodal nature of hyperspectral data poses significant challenges in accounting for all possible variations, making it difficult to develop a foundational model that is universally applicable to all hyperspectral datasets using existing models. The detailed characteristics of the available public datasets are shown in Table I.

3) *Lack of appropriate model architecture design:* Most current methods for HSI interpretation are typically designed for specific tasks within particular scenarios, often neglecting generalization performance in cross-scenario or cross-sensor contexts. Due to these design limitations, such models struggle to process hyperspectral data with varying dimensions. Recently proposed foundational models targeting cross-scenario and cross-sensor applications often trade off some spectral information in order to achieve broader generalization capabilities. Fig. 1 illustrates the challenges encountered in constructing foundation models for HSIs.

The analysis above suggests that developing a mechanism to uniformly represent this inherently variable modality could enable the construction of a large-scale foundational model compatible with all hyperspectral data. Some studies [35], [36] focused on spectral analysis have shown that the key spectral features indicative of material properties are primarily absorption or reflection peaks in the spectral signal. These peaks, along with variations in their position and intensity, sensitively reflect structural and compositional information. Additionally, we have observed certain similarities between spectral signals and natural language: both are sequential data types and both exhibit high contextual dependency [10]. Based on this theoretical foundation, we propose a new perspective for analyzing hyperspectral data: "**HyperS**pectral signals somewhat can be regarded as the "**L**anguage" of "natural world" that represents material properties."

Building on the aforementioned methodology, we have designed a large-scale foundational model specifically tailored for HSIs (**HyperSL**). The input layer of our model is a universal tokenizer, which is employed to convert spectral signals into unified token sequences by analogy to word embedding. Specifically, given a spectral signal $x$ where $x \in R^C$, the transformation is represented as $T : x \to \{t_1, t_2, \ldots, t_C\}$ where $t_i \in R^d$, and $d$ represents the embedding dimension of tokenizer. This representation naturally converts the dimensional variations in spectral signals from different platforms into differences in sequence length. Consequently, we can employ the transformer block as the backbone network to process spectral data of any dimension. Additionally, each dimension of the HSI corresponds to a specific spectral wavelength with distinct physical significance. The conventional sequential position encoding would disrupt this characteristic of the spectral data. To address this, we compute the position tokens based on the spectral wavelength information, preserving the physical details that would otherwise be lost during digitization. This design provides a unified spectral information scale for different types of HSIs that facilitates the development of a mechanism for the unified representation of the intrinsic multimodal nature of HSIs. Furthermore, considering the substantial information redundancy in spectral signals, we adopt the high masking ratio strategy from MAE that enables the model to efficiently extract contextual information from the spectral data. After pretraining, HyperSL allows the transfer of model knowledge to multiple downstream tasks without any modifications to the model. It is also applicable to hyperspectral data from various sensors, even if the model has not previously encountered data from these sensor types. We conducted experiments on two fundamental hyperspectral downstream tasks, classification and change detection, across multiple datasets. The experimental results demonstrate that HyperSL performs exceptionally well in both tasks. Our main contributions can be summarized as follows.

1) *Rethinking of spectral signals:* "Spectral signal can be perceived as the language of nature, somewhat can reveal the intrinsic properties of materials."
2) *Universal spectral tokenizer:* We transform spectral vectors of varying dimensions into a standardized token format, enabling a single model to process multisource hyperspectral data seamlessly.
3) *Spectral wavelength position encoding:* By incorporating wavelength information into positional encoding to retain the spectral frequency information of HSIs and enhance the alignment of spectral signals across different spectral ranges.
4) *Large-scale spectral pretraining:* Pretrained on over 300 million spectral instances from diverse global sources, our model captures a broad spectrum of wavelengths, bands, and environments, ensuring superior generalization.
5) *Effortless knowledge transfer:* Without requiring architectural modifications or training from scratch, our model seamlessly adapts to downstream tasks, establishing a universal framework for hyperspectral learning.

## II. Related Work

### A. Representation Learning for HSI

Representation learning has consistently played a crucial role in the interpretation of hyperspectral data. During the early stages of machine learning, many researchers used methods such as PCA [37], ICA [38], and LDA [39] to obtain low-dimensional representations of HSIs, thereby mitigating the dimensionality disaster and computational costs. Later, with the advent of deep learning, numerous researchers employed deep learning frameworks, including DNN [40], 1D-CNN [41], 2D-CNN [42], 3D-CNN [43], RNN [44], and Transformer [45], as backbone networks to extract features from hyperspectral data, generating more informative and representative feature vectors for downstream tasks such as classification, clustering, change detection, and unmixing [46]. In this period, another prominent representation method emerged based on autoencoders, where data is input into an encoder to obtain a feature vector, and then the feature vector is passed through a decoder to reconstruct the original data. After training, the decoder is discarded, and only the encoder

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serves as the backbone network, with an appropriate head appended for various downstream tasks [22]. For instance, Tao et al. [47] employed a stacked sparse autoencoder to learn spatial–spectral feature mappings of HSIs and then used SVM for classification. Yu et al. [48] used a spatial revising variational autoencoder for unsupervised feature extraction of HSIs and applied it for change detection. Gao et al. [49] coupled two sets of autoencoders as the backbone to simultaneously decompose input modalities into abundances and corresponding endmembers for unsupervised multispectral-aided HSI super-resolution. Although these methods achieved state-of-the-art performance in their respective tasks, they are mostly effective only within specific data scenarios. Once the data dimensions change, these models cannot perform inference without structural modifications and retraining, revealing significant limitations in scenario generalization performance. In our previous work [50], we made preliminary attempts to address this issue by cropping spectral data collected from different platforms and projecting them into a unified feature space via several downsampling convolutional layers, ensuring that the feature vectors have the same dimensionality. This small-scale pretrained model demonstrated a significant performance improvement in downstream classification tasks, also confirming the feasibility of training models using large-scale multisource hyperspectral data. However, this strategy achieves model generality at the cost of sacrificing some spectral information and does not account for the correlation between each dimension of the spectral vector and its actual wavelength. Additionally, the overly simplistic model design limits the capacity to accommodate extensive data knowledge, indicating room for further improvement.

### *B. Spectral RS Foundation Model*

Recently, the success of pretrained foundational models in natural language processing and computer vision has driven a series of studies on pretrained foundational models for spectral images, such as [18], [30], [33], [51], [52], [53], and [54]. These studies often employ self-supervised learning techniques to train a large-scale model on extensive unlabeled data, aiming to capture effective representations within the data. The two main self-supervised techniques used for spectral images are masked image modeling (MIM) and contrastive learning (CL). MIM seeks to reconstruct the original image or features from masked input images, as seen in models like MAE [31] and BEiT [55]. CL learns general representations by maximizing the similarity between augmented views of the same sample while minimizing it between different samples, as seen in methods like MoCo series [56], SimCLR [57], and Siamese network [58]. In particular, Wang et al. [59] proposed a nearest neighbor-based CL network (NNCNet) that takes full advantage of large amounts of unlabeled data to learn discriminative feature representations for multisource data classification. The use of MIM in foundational spectral image models is widespread due to the straightforward design of self-supervised tasks. For instance, Hong et al. [30] proposed the first large-scale foundational spectral model, SpectralGPT, which reconstructs spectral images in both spatial and spectral dimensions to enhance the model's learning of spectral sequences and employs a progressive pretraining strategy to integrate multiple datasets. Cong et al. [53] introduced SatMAE, a method that constructs a MIM autoencoder model capable of handling both temporal and spectral modalities, enhancing performance through independent masking across channels. Lin et al. [60] designed a spatial–spectral masked autoencoder (SS-MAE) for HSI and light detection and ranging (LiDAR)/synthetic aperture radar (SAR) data joint classification. It fully exploits the spatial and spectral representations of the input data, and both global and local features are taken into account for feature modeling. It employ PCA algorithm to unify the number of bands of multisource HSIs; however, different hyperspectral sensors capture data with varying spectral resolutions, wavelength ranges, and noise characteristics. Applying PCA separately to each dataset may result in different principal component distributions, making it difficult to maintain consistency across multisource images. Similarly, Wang et al. [33] proposed HyperSigma, a large-scale pretrained model for HSIs, which was tested on various downstream tasks. This model trains separately on spatial and spectral perspectives and aggregates them in the final stage, incorporating a sparse self-attention (SA) mechanism to accelerate training. It leverages multisource HSIs acquired from distinct sensors for pretraining, utilizing a random cropping technique to mitigate the dimensional discrepancies among HSIs. Nevertheless, this method compromises certain spectral details and substantially shifts the data distribution from real-world conditions. In contrast, SpectralEarth builds a foundational model using around 3.3TB of EnMap HSI data. However, its dataset lacks diversity and the model depends on a basic transformer architecture to enhance adaptability across various sensors. In our previous work [61], IMAE, we utilized input head rotation techniques to integrate different datasets for training the backbone network, designing an attention-guided mechanism to focus the model more on the central parts of selected patches. Work based on CL includes SkySense [32], a billion-scale multimodal remote sensing foundational model. SkySense uses a factorized multimodal spatiotemporal encoder to independently extract spatial features and perform multimodal temporal fusion, enabling support for multimodal remote sensing data. It constructs features from different modal and spatial granularities through multigranularity CL. While some foundational models accommodate various data formats, their implementation typically comes at the cost of information loss or constrained model scalability. For instance, to accommodate hyperspectral data with varying channels, HyperSigma randomly selects the same number of consecutive channels within hyperspectral datasets as input data, which compromises the true data distribution and results in a loss of spectral information. IMAE and SatMAE accept specified datasets through different input heads for pretraining the backbone model, offering limited flexibility during deployment; new datasets often require custom input heads and retraining of the model. In contrast, HyperSL, after pretraining, can transfer the learned information to any dataset without modifying the

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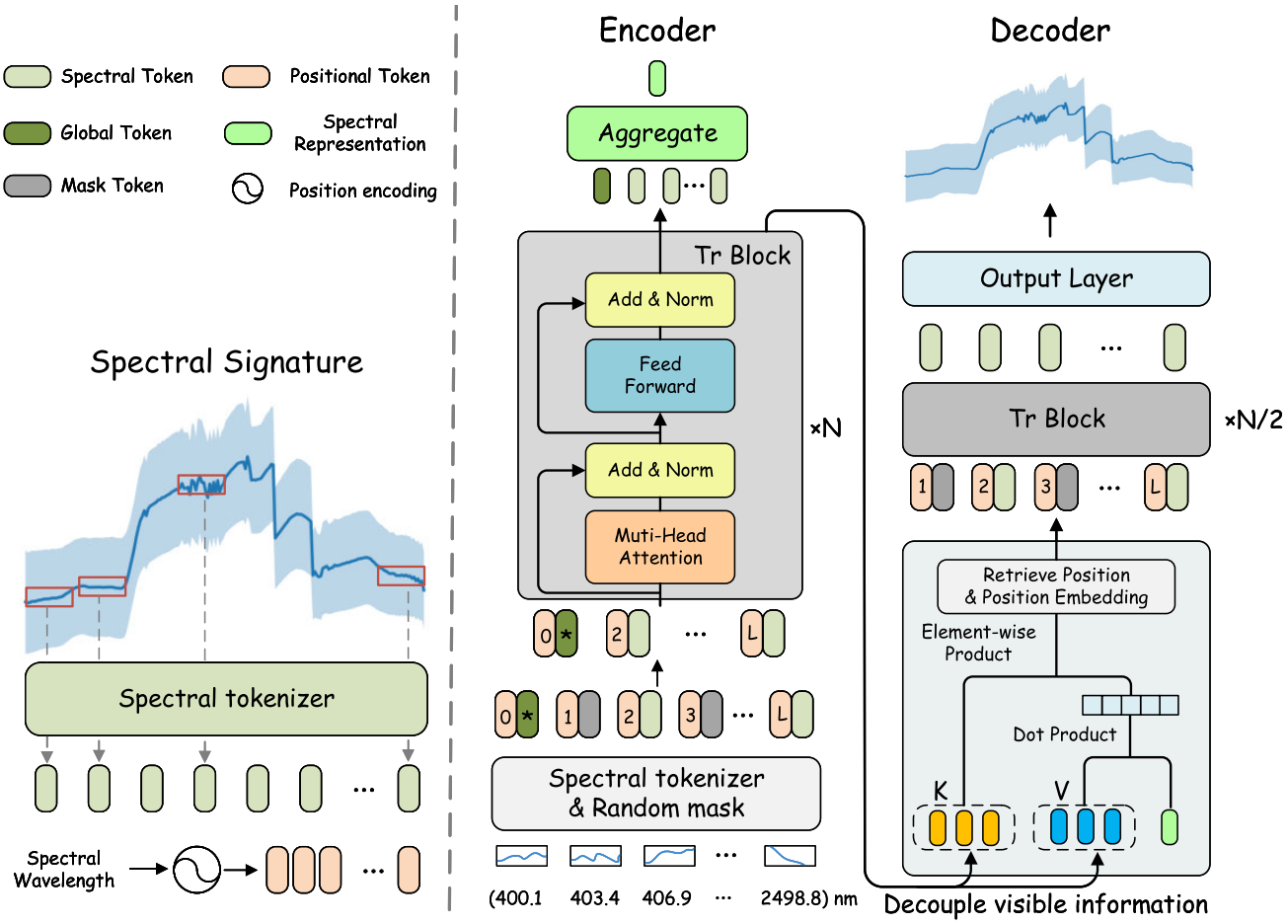



Fig. 2. Overall structure of HyperSL. First, the position tokens computed based on its spectral wavelength for each band. Then, the spectral signals are projected into the token sequences by the spectral tokenizer, and add the position tokens to them. Finally, we apply a high-ratio random masking to the spectral token sequences containing wavelength information and use an autoencoder to reconstruct the input sequences back to the original spectral signals.

model, even if it has never encountered data from such sensor types before.

## III. Methodology

HyperSL is primarily composed of the following components: a universal spectral tokenizer, spectral wavelength embedding, and a spectral masked generation model. First, we compute the position token for each spectral band according to its spectral wavelength. Next, we use a spectral tokenizer to convert the spectral signals into the sequences of tokens, which is then combined with the position tokens. Finally, we apply a high-proportion random masking to each spectral token sequence and use an autoencoder to reconstruct the input sequence into the original spectral signals. The structure of HyperSL is illustrated in Fig. 2, and the specific technical details will be elaborated in the subsequent sections.

### A. Universal Spectral Tokenizer

Hyperspectral images (HSIs) from different platforms often have varying numbers of spectral channels. Traditional models, once trained, are typically limited to processing data with fixed shapes. As a result, specific model architectures are usually designed for particular datasets, or methods are employed that discard some spectral information to standardize the data shapes. The first approach restricts the model to a single data source, limiting the effective use of available data and reducing the diversity of features learned. The second approach introduces significant discrepancies between the training data and the actual data distribution.

To address these challenges, we developed a spectral tokenizer that projects all spectral signals into sequence-like representations, akin to word vectors, by converting variations in the number of spectral channels into differences in sequence length. Furthermore, considering that spectral signals exhibit varying resolutions and ranges, we employ multiscale windows for segmentation, allowing the model to accommodate spectral data with diverse distributions. Specifically, let the spectral signal be $x \in \mathbb{R}^c$, where $c$ represents the spectral bands of the input HSI. We first use multiscale 1-D convolutions to project it

$$x^{(1)} = \text{Conv1D}(x), \quad x^{(1)} \in \mathbb{R}^{n\times c} \tag{1}$$
$$x^{(2)} = \text{Conv1D}(x), \quad x^{(2)} \in \mathbb{R}^{n\times c} \tag{2}$$
$$x^{(3)} = \text{Conv1D}(x), \quad x^{(3)} \in \mathbb{R}^{n\times c} \tag{3}$$

where $n$ represents channel numbers of the output features, and it can be manually specified. Then, fuse the obtained multiscale features along the last dimension and then

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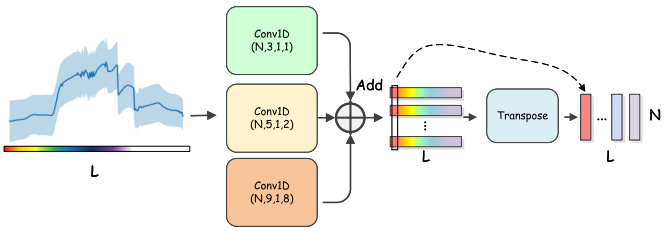



Fig. 3. Architecture of universal spectral tokenizer.

transpose them

$$x^{(4)} = \left(\text{Conv1D}\left(x^{(1)} + x^{(2)} + x^{(3)}\right)\right)^{\top}, \quad x^{(4)} \in \mathbb{R}^{c\times n}. \quad (4)$$

In this way, we obtain spectral tokens where each token implicitly embeds the spectral wavelength information and its context. Moreover, spectral differences are converted into variations in sequence length. This transformation enables the use of transformer modules as the backbone for constructing a universal model. Fig. 3 shows the architecture of the universal spectral tokenizer.

### B. Spectral Wavelength Position Encoding

Each channel of HSI represents the reflectance intensity at a specific spectral wavelength. In remote sensing products, these wavelength details are typically stored in a separate metadata file. However, raster files that store HSIs lack this information. Current methods only analyze HSIs using raster files, ignoring the wavelength-channel relationship. Transformer-based methods for HSI interpretation typically compute position tokens based solely on the natural order of the channels. This approach is feasible for data from a specific platform, as it effectively serializes the wavelength information. While this approach is effective for data from a specific platform, it only encodes the wavelength information within a fixed order. However, when processing different hyperspectral data, datasets vary in wavelength ranges and spectral resolutions. Therefore, relying solely on the natural order of the channels for position encoding is inadequate. To bridge this gap, we propose a position encoding strategy that integrates spectral wavelength information. By associating each HSI channel with its corresponding physical wavelength characteristics, this strategy ensures broad applicability across diverse hyperspectral datasets. Specifically, let $w_i \in \mathbb{R}$ denote the spectral wavelength corresponding to the $i$th channel of the HSI. The mathematical expressions for calculating the position token are provided below

$$\text{PE}(w_i, 2j+1) = \cos\left(\frac{w_i - 400}{10\,000^{2j/d}}\right) \quad (5)$$

$$\text{PE}(w_i, 2j) = \sin\left(\frac{w_i - 400}{10\,000^{2j/d}}\right) \quad (6)$$

where $d$ represents the dimensionality of the position token, $j$ denotes the $j$th dimension of this vector, and 400 is the bias of spectral wavelength. By implementing this strategy, each channel associated with a specific wavelength is assigned a unique position token, and the position tokens of adjacent channels maintain continuity. Figs. 4 and 5 visualize the spectral position tokens for ENMap and DESIS, respectively.

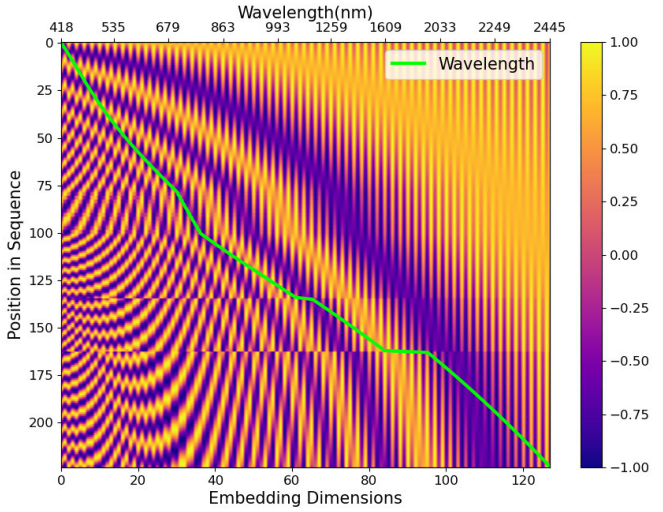



Fig. 4. Visualization of position tokens for ENMap. The vertical axis represents the position in the sequence, the top horizontal axis represents the spectral wavelength, the bottom horizontal axis represents different encoding dimensions, and the green line represents the correspondence between the spectral wavelength and the sequence position. Each spectral signal at each position is unique and maintains a linear correlation with the spectral wavelength. It is important to note that the nonlinear segments in the figure are due to the removal of bad bands from the ENMap data.

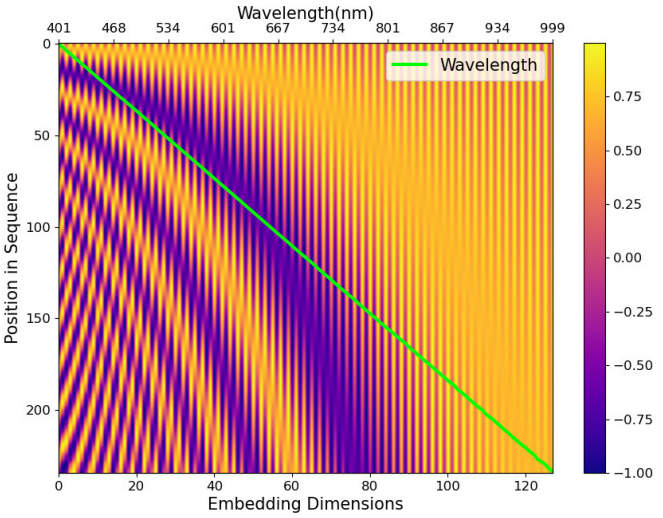



Fig. 5. Visualization of position tokens for DESIS.

### C. Spectral Masked Generation Model

MIM models have demonstrated great promise in image representation. High-proportion masking strategies are commonly employed in MIM methods for diverse images to reduce redundancy in image data and improve feature learning. As stated in MAE: *"Information density is different between language and vision."* Such a phenomenon is also observed in hyperspectral signals, as numerous studies indicate that hyperspectral data are highly redundant. During the tokenization of spectral signals, we introduce a substantial amount of overlapping contextual information that exacerbates the redundancy. Drawing inspiration from the concept of MAE, we adopt a high masking ratio strategy to reduce redundancy in the signals. This strategy increases the difficulty of the model's reconstruction task, forcing the model to capture higher-level semantic information within the spectral signals. The masked

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generation module includes a Transformer-based encoder and a small decoder.

On the encoder side, let $x \in \mathbb{R}^c$ denote the input hyperspectral signal. It is encoded into a token sequence $\{t_1, \ldots, t_n\}$, where $t_i \in \mathbb{R}^n$, using a multiscale tokenizer. The position tokens $\{p_i\}$, $p_i \in \mathbb{R}^n$ are calculated from their central wavelength information $\{w_i\}, w_i \in \mathbb{R}$. First, we combine the position encodings with the spectral tokens to obtain the input sequence $H = \{h_i\}$ for the encoder

$$H = T + P. \tag{7}$$

Next, we apply a high-ratio random masking to these tokens and concatenate a CLS token to the visible tokens. The visible tokens $H_{\text{vis}} = \{\text{CLS}, h_m\}$ are input into an encoder composed of multiple Transformer modules for processing. The computation process is as follows:

$$H_v^{(l)'} = \text{MSA}\left(\text{LN}\left(H_v^{(l-1)}\right)\right) + H_v^{(l-1)} \tag{8}$$

$$H_v^{(l)} = \text{LN}\left(\text{MLP}\left(H_v^{(l)'}\right)\right) + H_v^{(l)'}. \tag{9}$$

Here, MSA denotes multihead SA, which is formed by concatenating multiple SA layers

$$\text{MSA}(H_v) = [\text{SA}_1(H_v), \text{SA}_2(H_v), \ldots, \text{SA}_n(H_v)]W. \tag{10}$$

SA is computed using three learnable matrices $W_Q$, $W_K$, and $W_V$, which map the input tokens to their own Query ($Q$), Key ($K$), and Value ($V$)

$$Q = W_Q H_v, \quad K = W_K H_v, \quad V = W_V H_v. \tag{11}$$

Each SA layer is calculated as

$$\text{SA}(H_v) = \text{softmax}\left(\frac{QK^\top}{\sqrt{D}}\right)V. \tag{12}$$

At the output layer of the encoder, we aggregate all visible tokens and the CLS token to form the spectral signal representation vector, and output the corresponding $K$ and $V$ for all visible tokens.

The decoder consists of a lightweight Transformer module. At its input, we utilize the $K$ and $V$ outputs from the encoder to decouple the information of visible bands from the representation vector

$$H_v = K \cdot H_v^\top \cdot V. \tag{13}$$

We then restore them to their original positions before shuffling, filling the positions of the masked tokens with randomly initialized mask tokens, and adding positional information to all tokens. At the output layer of the decoder, we reconstruct the token sequence back to the input spectral signals.

We use a simple mean squared error (mse) as the loss function to optimize the network. Since spectral signals are vector signals, their directional characteristics are as important as their magnitudes. To explicitly incorporate this aspect, we introduce an additional constraint on the vector direction by utilizing cosine similarity. The MSE term minimizes the absolute reconstruction error, ensuring that the magnitude of the reconstructed signal approximates the original signal. Meanwhile, the cosine similarity term minimizes the angular difference between the reconstructed and the original signal, thereby preserving the directional consistency of the spectral signal. To prevent overfitting, we add $L_2$ regularization to the network. The network is optimized according to the following formula:

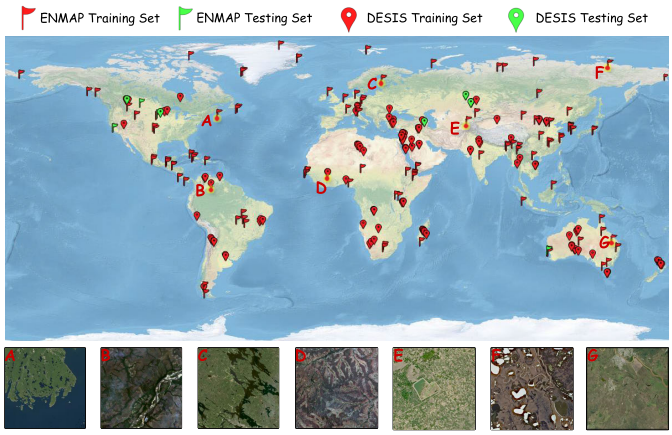



Fig. 6. General geographical distribution of the pretraining dataset. The dataset is collected from hyperspectral remote sensing images provided by ENMap and DESIS from various locations worldwide, encompassing diverse scenes such as oceans, wetlands, mountains, permafrost, rivers, farmlands, rural areas, and urban environments. The overall cloud coverage of the data is below 10%. In the figure, data partitioning is annotated using red-green legends, with the test set primarily used to monitor convergence during the pretraining process, allowing timely adjustments to the training schedule.

$$\Theta^* = \arg\min_\theta \left(\text{MSE}(x, \hat{x}) + \text{CosSim}(x, \hat{x}) + \alpha\|\theta\|\right) \tag{14}$$

where $\Theta^*$ represents the optimal weights of the model, $\hat{x}$ represents the reconstructed spectral signal, $\text{CosSim}(x, \hat{x})$ denotes the cosine similarity between $x$ and $\hat{x}$ that enforces the directional alignment, and $\|\cdot\|$ is the $L_2$ norm of the network parameters. The regularization coefficient $\alpha$ controls the strength of the regularization term.

## IV. Experiments

### A. Pretraining

In this section, we will provide a detailed explanation of the implementation details during the pretraining phase and discuss the experimental results.

*1) Dataset Description:* Our dataset primarily utilizes spaceborne hyperspectral data from two types of sensors: EnMap and DESIS. The environmental mapping and analysis program (EnMAP) provides unique data essential for tackling significant environmental challenges related to human activities and climate change. It utilizes a dedicated pushbroom HSIr, primarily based on modified existing or predeveloped technology. EnMAP covers a broad spectral range from 420 to 1000 nm in the visible and near-infrared (VNIR) region, and from 900 to 2450 nm in the shortwave infrared (SWIR) region, with high radiometric resolution and stability across both ranges. Additionally, it offers a swath width of 30 km with a spatial resolution of 30 × 30 m. The hyperspectral instrument, DLR Earth sensing imaging spectrometer (DESIS), is one of four payloads of the multiuser system for Earth sensing (MUSES), mounted on the International Space Station (ISS). Operating in the wavelength range from visible to near-infrared, DESIS enables precise data acquisition of Earth's surface for various applications, including fire

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detection, change monitoring, maritime domain awareness, and atmospheric research. It provides a spatial resolution of approximately 30 m on the ground and is sensitive to wavelengths between 400 and 1000 nm, with a spectral resolution of about 3.3 nm. DESIS data are delivered in tiles covering approximately 30 × 30 km. Both EnMap and DESIS data are sourced from the DLR GeoPortal. For these datasets, we selected data from various regions around the world across different seasons, incorporating a range of common land-cover types, with cloud cover near zero, totaling approximately 200 million spectral data points. The general geographical distribution is shown in Fig. 6. For EnMap, we removed the bad bands [128–134]. For DESIS, we removed the bad bands [1–6]. Additionally, we used airborne hyperspectral data provided by the Shanghai Bureau of Surveying and Mapping for pretraining. The data spans a spectral range of 400–2500 nm, with the VNIR spectral resolution at 2.5 nm and the SWIR spectral resolution at 7 nm, comprising a total of 439 bands. It has a spatial resolution of 1.5 m. To ensure that the features of each band are on the same scale, we performed layer-by-layer normalization on these HSI datasets.

*2) Implement Details:* To accelerate data loading during training, we stored the data from the two satellites in HDF5 format. Due to inconsistent spectral dimensions across different data sources, it is impossible to load them into the same batch for training in the output model. To prevent data bias in the model, we employed two separate data loaders for the two data sources. After the gradient updates from both loaders are completed, they are cumulatively updated together.

In the mask generation model, the mask rate is a critical hyperparameter, as it significantly impacts the difficulty of the reconstruction task and the model's computational efficiency. A high mask rate means an extremely challenging reconstruction task, potentially leading the model to learn underlying mechanisms within the data representation, whereas a low mask rate may prevent adequate model training due to overly simple reconstruction tasks. Since our proposed universal tokenizer uses overlapping calculations to obtain spectral token sequences, resulting in high information redundancy, we chose a high mask rate of 95% for the generated sequence to ensure sufficient training for the model. The choice of a 95% mask rate is based on the intuition that, in the implementation, the largest 1-D convolution kernel used by the tokenizer is 13, while most hyperspectral data dimensions are around 200. Thus, in a non-overlapping situation, a kernel size of 13 can represent the spectral data in about 15 tokens. A 95% mask rate ensures approximately ten visible tokens per instance, yielding an information amount equivalent to a 40% mask rate on the original spectrum in the highest case (uniform sampling) and about 85% in the lowest case (sampling the rightmost ten tokens). The 40%–85% range is suitable for most reconstruction tasks, and applying a 95% mask rate on the spectral token sequence greatly reduces the computational cost of model training. For pretraining, we used the AdamW optimizer with a weight decay of 0.0005. CosineAnnealingWarmRestarts was applied to adjust the learning rate, with an initial T_0 of 10, T_mult of 2, and an initial learning rate of 0.0005. The training was conducted on three A800 GPUs.

TABLE II
PRETRAINING COST FOR THE HYPERSL BASE MODEL

| Model | Embed dim | Param.(M) | FLPOS(G) | Time |
|---|---|---|---|---|
| HyperSL-small | 128 | 0.23 | 0.53 | 37h |
| HyperSL-base | 256 | 9.03 | 2.12 | 125h |

* The FLOPS is calculated on a single training sample and the time consumption is recorded under full load of the GPUs (3 A800 80GB).

TABLE III
DETAILED INFORMATION OF THE DATASETS USED

| Datasets | Methods | MSE↓ | SSIM↑ | PSNR/dB↑ | SAM↓ |
|---|---|---|---|---|---|
| IP | WL PE | 0.01 | 0.8961 | 19.81 | 8.43 |
| | Seq PE | 0.027 | 0.7512 | 15.69 | 20.21 |
| BOS | WL PE | 0.031 | 0.7811 | 16.09 | 5.67 |
| | Seq PE | 0.045 | 0.6528 | 13.45 | 16.13 |

* WL PE represents wavelength position encoding and Seq PE represents sequential position encoding.

The preset pretraining epoch was 50; however, the model reached convergence around the tenth epoch, with a training time of approximately 12.5 h per epoch. To evaluate the model's reconstruction capability on other datasets, we used MSE, SAM, PSNR, and SSIM as evaluation metrics. The model's parameters and computational complexity are shown in Table II.

*3) Results and Ablation Analysis:* After the pretraining phase, to evaluate the model's reconstruction capability in cross-sensor scenarios, we selected the widely used Indian Pines (IP) dataset collected from AVIRIS and the Botswana (BOS) dataset from EO-1 as test data. The data from these two sensors is entirely anonymous to the model. Furthermore, to investigate the impact of spectral position encoding on cross-sensor reconstruction, we conducted an ablation analysis on the two unseen datasets with different spectral bands. In the ablation experiments, we replaced the spectral encoding with sequential encoding. The experimental results are presented in Table III. The results indicate that the reconstruction accuracy achieved using the spectral position encoding network is significantly higher than that of sequential position encoding, particularly in terms of the SAM. This suggests that spectral position encoding provides HyperSL with additional parametric information to align distribution discrepancies between multisource HSIs, enabling the model to robustly learn generalizable features in cross-source scenarios. Moreover, we randomly sampled four pixels in the BOS and IP datasets, and their spectral reconstruction curves are shown in Figs. 7 and 8. As seen in the figure, when wavelength encoding is used, the spectral curve envelope is effectively reconstructed, whereas with sequential encoding, the reconstructed spectral curves exhibit significant distortion. It indicates that HyperSL is capable of effectively reconstructing spectral features even from previously unseen sensor data. The reconstruction outcomes utilizing spectral encoding significantly outperform those achieved with sequential positional encoding across all evaluation metrics. Notably, in terms of spectral angle, the

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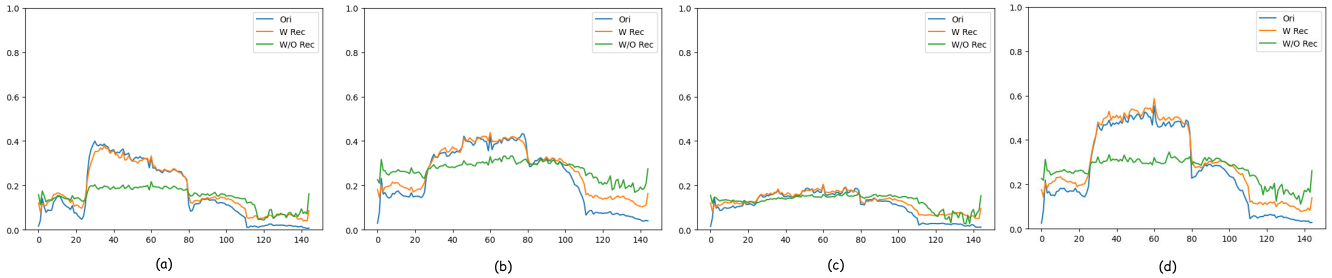



Fig. 7. Spectral reconstruction results of BOS. "Ori" refers to the original spectral curve, "W/O Rec" refers to the spectral curve reconstructed without spectral position encoding, and "W Rec" refers to the spectral curve reconstructed with spectral position encoding. (a) Spectral curves of point (100, 100). (b) Spectral curves of point (100, 200). (c) Spectral curves of point (200, 100). (d) Spectral curves of point (200, 200).

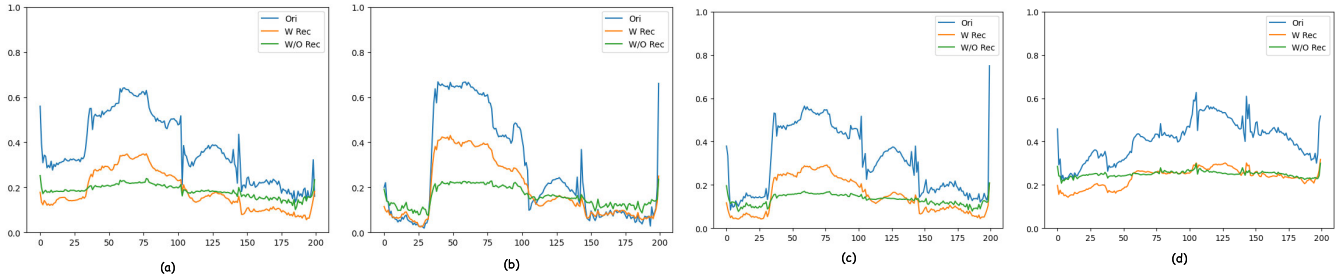



Fig. 8. Spectral reconstruction results of IP. "Ori" refers to the original spectral curve, "W/O Rec" refers to the spectral curve reconstructed without spectral position encoding, and "W Rec" refers to the spectral curve reconstructed with spectral position encoding. (a) Spectral curves of point (50, 50). (b) Spectral curves of point (50, 100). (c) Spectral curves of point (100, 50). (d) Spectral curves of point (100, 100).

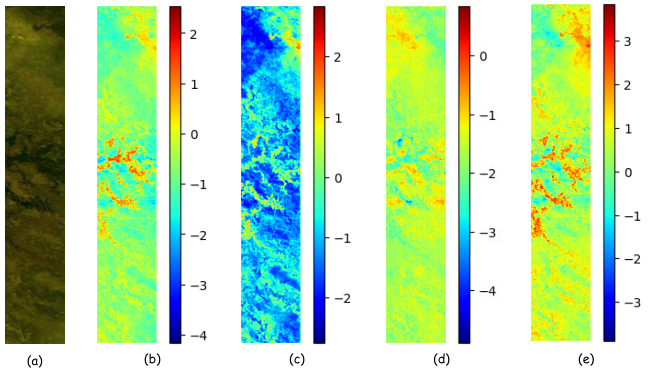



Fig. 9. Feature visualization of the BOS dataset. (a) False RGB image. (b) Channel 1 of the feature representation. (c) Channel 145 of the feature representation. (d) Channel 216 of the feature representation. (e) Channel 254 of the feature representation.

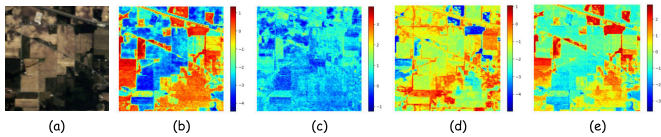



Fig. 10. Feature visualization of the IP dataset. (a) False RGB image. (b) Channel 1 of the feature representation. (c) Channel 145 of the feature representation. (d) Channel 216 of the feature representation. (e) Channel 254 of the feature representation.

reconstruction results with spectral positional encoding show differences of 11.78°–10.46° compared to sequential positional encoding for the IP and BOS datasets, respectively. This highlights the critical role that spectral positional encoding plays in enhancing cross-sensor capabilities.

Furthermore, we conducted a visual analysis of the features output by the HyperSL encoder to explore the information learned by the model at the semantic level. Figs. 9 and 10 illustrate the feature visualizations for channels 1, 145, 216, and 254 of the BOS and IP datasets, respectively. For channel 1, the model shows a strong response in areas with high vegetation coverage in the IP data, while it exhibits a lower response in more barren land. In the BOS dataset, the model responds strongly to water bodies, shows a moderate response in vegetated areas, and has a lower response for exposed surfaces like riverbanks. For channel 145, the model demonstrates a generally low response across the IP dataset, with relatively higher responses in vegetated regions; similarly, it maintains a low overall response in the BOS dataset, with a notable increase only in water-covered areas. In channel 216, the model exhibits moderate responses in barren land areas within the IP dataset, while differentiating its response for various vegetation types—showing low responses for crops and trees, but higher responses for wheat and grasslands. On the BOS dataset, characterized by its wetland environment and dense vegetation coverage, the model exhibits an overall moderate response, with lower responses observed in areas of exposed soil. For channel 254, the model's responses are analogous to those in channel 1; it exhibits high responses in vegetated areas of the IP dataset and lower responses in barren land, though slightly higher than in channel 1. In the BOS dataset, the model again responds strongly to water bodies, with moderate responses in vegetated areas, also slightly higher than in channel 1, and a lower response for exposed surfaces. This indicates that the model has learned certain material characteristics at the semantic level, such as a clear representation of the spatial distribution of moisture content in observed land cover and spectral semantic characteristics of vegetation, enabling it to distinguish between types like trees and grasslands. Notably, this analysis is based on two completely anonymous hyperspectral datasets from different sensors, which exhibit similar trends in feature representation,

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TABLE IV
DETAILED INFORMATION OF IP DATASET

| No. | Class Name | Training samples | Test samples | Total samples |
|---|---|---|---|---|
| 1 | Water | 27 | 243 | 270 |
| 2 | Hippo grass | 11 | 90 | 101 |
| 3 | Floodplain grasses 1 | 26 | 225 | 251 |
| 4 | Floodplain grasses 2 | 22 | 193 | 215 |
| 5 | Reeds | 27 | 242 | 269 |
| 6 | Riparian | 27 | 242 | 269 |
| 7 | Firescar | 26 | 233 | 259 |
| 8 | Island interior | 21 | 182 | 203 |
| 9 | Acacia woodlands | 32 | 282 | 314 |
| 10 | Acacia shrubs | 25 | 223 | 248 |
| 11 | Acacia grasslands | 31 | 274 | 305 |
| 12 | Short mopane | 19 | 162 | 181 |
| 13 | Mixed mopane | 27 | 241 | 268 |
| 14 | Bare soil | 10 | 85 | 95 |
| | Total | 331 | 2917 | 3248 |

providing strong evidence for the model's generalization capabilities.

### B. Classification

Classification is a fundamental task in hyperspectral interpretation. In this section, we fine-tune the pretrained model on several commonly used classification datasets and conduct ablation analysis on the effectiveness of pretrained weights and the impact of spectral position encoding on classification accuracy.

*1) Dataset Description:* We selected the BOS and IP datasets as benchmark datasets for classification tasks. BOS dataset was acquired by NASA EO-1 satellite over the Okavango Delta in BOS from 2001 to 2004. The Hyperion sensor on EO-1 captures data at a 30-m pixel resolution across a 7.7-km strip in 242 bands spanning the 400–2500 nm range with 10 nm spectral windows. The UT Center for Space Research performed data preprocessing to mitigate the effects of faulty detectors, inter-detector miscalibration, and intermittent anomalies. Uncalibrated and noisy bands covering water absorption features were removed, leaving 145 bands as candidate features: [10–55, 82–97, 102–119, 134–164, 187–220]. The data analyzed in this study, collected on May 31, 2001, include observations from 14 identified classes representing land-cover types in seasonal swamps, occasional swamps, and drier woodlands located in the distal parts of the Delta. The ground truth provides 16 different land-cover classes, as shown in Fig. 11, with the number of samples for each class listed in Table IV. The wavelength information was obtained from the EO-1 data meta files available through Portal Gscloud. The IP dataset was collected by the AVIRIS sensor over the IPs test site in northwestern Indiana and comprises 145 × 145 pixels with 224 spectral reflectance bands in the wavelength range of 0.4–2.5 $\mu$m. The number of bands was reduced to 200 by removing water absorption bands: [104–108], [150–163], and 220. The ground truth provides 16 different land-cover classes, as shown in Fig. 12, with the number of samples for each class listed in Table V. The wavelength information can be accessed from Purdue University's MultiSpec site.

*2) Implement Details:* When fine-tuning the pretrained model for classification tasks, we use only the encoder part of HyperSL. First, patches of hyperspectral data to be classified

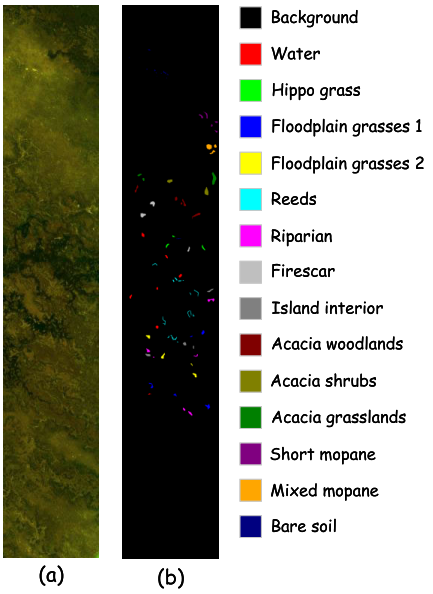



Fig. 11. BOS dataset. (a) Pseudocolor image. (b) Groundtruth.

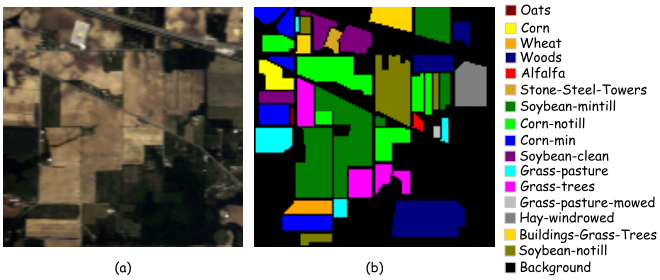



Fig. 12. IP dataset. (a) Pseudocolor image. (b) Groundtruth.

TABLE V
DETAILED INFORMATION OF BOS DATASET

| No. | Class Name | Training samples | Test samples | Total samples |
|---|---|---|---|---|
| 1 | Alfalfa | 5 | 41 | 46 |
| 2 | Corn-notill | 143 | 1285 | 1428 |
| 3 | Corn-min | 83 | 747 | 830 |
| 4 | Corn | 24 | 213 | 237 |
| 5 | Grass-pasture | 49 | 434 | 483 |
| 6 | Grass-trees | 73 | 657 | 730 |
| 7 | Grass-pasture-mowed | 3 | 25 | 28 |
| 8 | Hay-windrowed | 48 | 430 | 478 |
| 9 | Oats | 2 | 18 | 20 |
| 10 | Soybean-notill | 98 | 874 | 972 |
| 11 | Soybean-mintill | 246 | 2209 | 2455 |
| 12 | Soybean-clean | 60 | 533 | 593 |
| 13 | Wheat | 21 | 184 | 205 |
| 14 | Woods | 127 | 1138 | 1265 |
| 15 | Buildings-Grass-Trees | 39 | 347 | 386 |
| 16 | Stone-Steel-Towers | 10 | 83 | 93 |
| | Total | 1031 | 9218 | 10249 |

are flattened and uniformly represented using HyperSL. Let the sampled HSI patch be $X \in R^{p\times p\times C}$, where $p$ represents the patch size and $C$ is the number of channels in the HSI. Specifically, we transform $X$ to $X_f \in R^{p^2\times 1\times C}$, adding an extra dimension to facilitate 1D-CNN operations. Next, $X_f$ is input into HyperSL

$$H_f = \text{HyperSL_encoder}(X_f) \tag{15}$$

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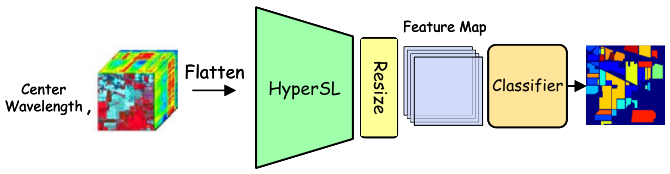



Fig. 13. Overall architecture of HyperSL for classification task.

where $H_f \in R^{p^2 \times 1 \times d}$, and $d$ is the dimensionality of the output representation vector, which can be user-specified. Then, we resize the features into a feature map and use a simple convolutional classifier for classification. Transform $H_f$ to $H \in R^{p \times p \times d}$, and pass $H$ through a convolutional module and a global average pooling module to obtain the feature vector $H'$

$$H' = \text{GAP}(\text{Conv Block}(H)). \tag{16}$$

Next, an MLP is used to generate the classification logits for the data

$$\text{logits} = \text{MLP}(H'). \tag{17}$$

Finally, cross-entropy loss is applied for end-to-end fine-tuning optimization of the model

$$\theta^* = \arg\min_{\theta} \text{CrossEntropy}(\text{logits}, y). \tag{18}$$

The overall architecture of classification network is illustrated in Fig. 13.

In the experiments, we set the patch size to five both for the IN and BS datasets. In each category, 10% of the samples are allocated as the training set, while the remaining samples serve as the test set. Detailed information about the dataset is provided in Table V. The model is trained for a total of 150 epochs, with the learning rate set to 0.0001, cross-entropy as the loss function, and the AdamW optimizer applied with a weight decay of 0.05. Classification performance is evaluated using OA, AA, and Kappa as metrics. We compare our approach with current mainstream and advanced classification methods, including the following: CVT [62], Wave-Vit [63], RVT [64], CLOLN [65], HSIC-FM [66], CSIL [67], Spectralformer [45], SS-MTr [68], HyperSigma [33]. CVT, Wave-Vit, and RVT are advanced vision transformers, which we have adapted for hyperspectral data. CLOLN, HSIC-FM, CSIL, and Spectralformer are network architectures meticulously designed to exploit the unique characteristics of hyperspectral data, with their primary structures built on Transformer and attention mechanisms. SS-MTr and HyperSigma employ pretrained-based training strategies, where SS-MTr learns feature representations on target datasets through masked reconstruction, while HyperSigma captures feature representations by leveraging extensive hyperspectral data.

*3) Results Analysis:* Tables VI and VII present the classification results of different methods on the BOS and IP datasets, respectively. The classification result maps are shown in Figs. 14 and 15, respectively. Our approach achieves state-of-the-art performance across all metrics compared with other methods. For the BOS dataset, our method achieves OA, AA, and Kappa scores of 99.66%, 99.67%, and 99.63%, respectively, outperforming all mainstream approaches. For the IP dataset, our method achieves OA, AA, and Kappa scores of 98.48%, 98.70%, and 98.27%, respectively. Among these methods, our approach achieved the best performance in terms of OA and AA, while the Kappa score was slightly inferior to HyperSigma by a margin of 0.03%, securing the second-best result. For both datasets, we observe that methods trained from scratch consistently underperform compared to pretrained-based approaches. This is primarily due to the inability of models to adequately learn representations under limited sample conditions. Among the pretrained methods, SS-MTr pretrains exclusively on the target dataset, yielding favorable performance on the specific dataset but demonstrating limited generalization ability, as it requires pretraining anew for each dataset. HyperSigma, leveraging large-scale data for pretraining, achieves outstanding performance in downstream tasks. However, its spectral channel processing relies on random sampling, necessitating extensive data for training. This results in high data requirements and significant training costs. In contrast, our method employs a unified tokenizer to convert spectral signals into token sequences while incorporating spectral wavelength encoding, embedding the actual wavelength information of the spectral signals. This establishes a standardized positional encoding scheme, enabling the model to learn transferable spectral representations. Consequently, the pretraining cost is reduced. During fine-tuning, the shared positional encoding standard between pretrained and target datasets facilitates feature alignment, thereby enhancing transferability and yielding superior performance.

Moreover, we perform ablation experiments using randomly initialized network weights to analyze the impact of pretraining on improving downstream classification tasks. The randomly initialized HyperSL achieves OA, AA, and Kappa scores of 97.05%, 97.20%, and 96.81% on the BOS dataset, and 95.04%, 95.51%, and 94.34% on the IP dataset, respectively. By initializing the model with pretrained weights, the performance on the BOS dataset improves by 2.61%, 2.47%, and 2.82% for OA, AA, and Kappa, respectively, while on the IP dataset, the improvements are 3.44%, 3.19%, and 3.93% for the same metrics. These results indicate that pretraining effectively enhances performance on downstream classification tasks.

To evaluate the stability and robustness of the model, we conducted ten independent experiments on two datasets using different proportions of training samples, as illustrated in Fig. 16. For the BOS dataset, with only 2% of the samples, it achieves mean OA, AA, and Kappa scores of 89.33%, 89.99%, and 88.46%, respectively, though with relatively high standard deviations of 5.08%, 4.63%, and 8.45%. When the training proportion increases to 5%, the model's accuracy improves substantially, reaching 98.04% OA, 97.98% AA, and 97.88% Kappa, while the standard deviations decrease to 1.25%, 1.11%, and 1.35%, reflecting greater stability. With 10% of the training samples, the model further enhances its performance, achieving 99.66% OA, 99.67% AA, and 99.63% Kappa, with standard deviations reduced to 0.22%, 0.19%, and 0.24%, demonstrating high consistency. Finally, when trained with 15% of the samples, the model nearly reaches saturation,

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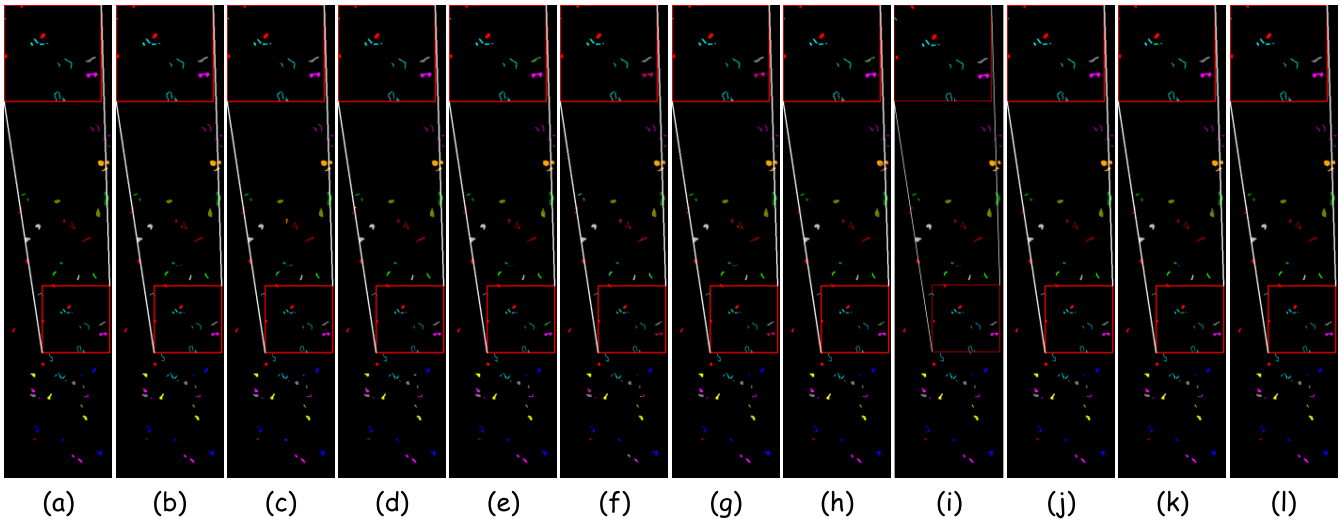



Fig. 14. Classification maps for the BOS dataset. (a) Groundtruth. (b) CVT. (c) Wave-Vit. (d) RVT. (e) CLOLN. (f) HSIC-FM. (g) CSIL. (h) Spectralformer. (i) SS-MTr. (j) HyperSigma. (k) W/O HyperSL. (l) HyperSL. W/O represents without pretrained.

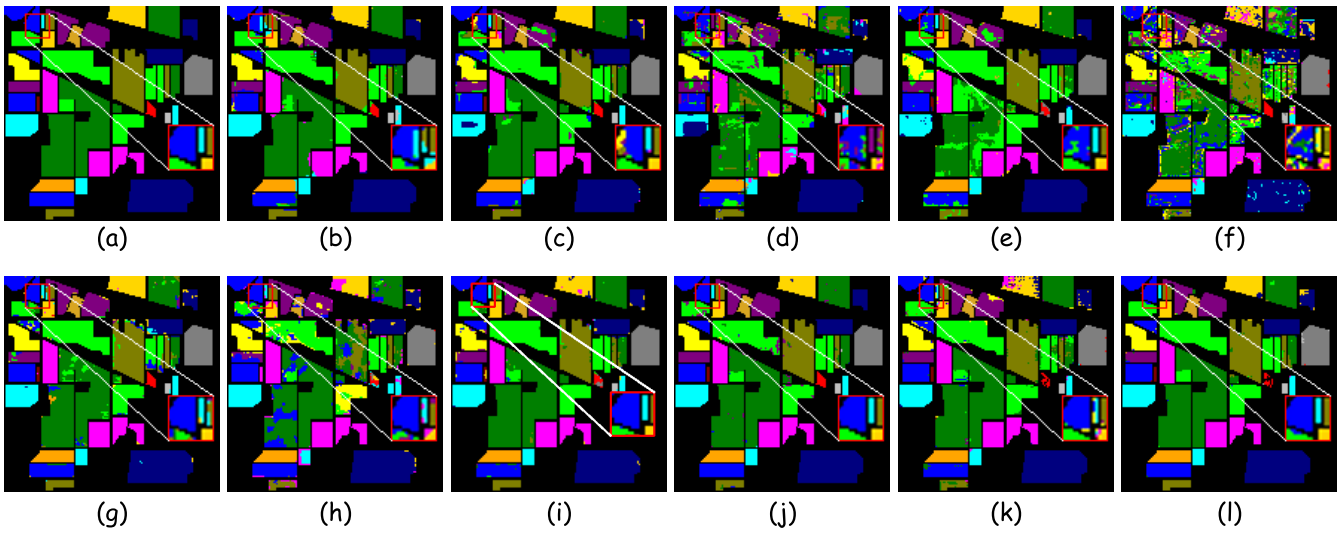



Fig. 15. Classification maps for the IP dataset. (a) Groundtruth. (b) CVT. (c) Wave-Vit. (d) RVT. (e) CLOLN. (f) HSIC-FM. (g) CSIL. (h) Spectralformer. (i) SS-MTr. (j) HyperSigma. (k) W/O HyperSL. (l) HyperSL. W/O represents without pretrained.

TABLE VI

CLASSIFICATION RESULTS OF DIFFERENT METHODS USING 10% TRAINING SAMPLES PER CLASS ON BOS DATASET

| Classes | CVT [62] | Wave-Vit [63] | RVT [64] | CLOLN [65] | HSIC-FM [66] | CSIL [67] | SPEC-F [45] | SS-MTr [68] | HyperSigma [33] | W/O HyperSL | HyperSL |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 98.15% | 98.52% | **100.00%** | **100.00%** | **100.00%** | **100.00%** | **100.00%** | **100.00%** | **100.00%** | **100.00%** | **100.00%** |
| 2 | **100.00%** | 93.07% | 99.01% | 98.06% | 68.13% | **100.00%** | **100.00%** | **100.00%** | **100.00%** | **100.00%** | **100.00%** |
| 3 | **100.00%** | 97.61% | 93.63% | **100.00%** | 94.47% | 99.12% | 99.12% | **100.00%** | **100.00%** | **100.00%** | **100.00%** |
| 4 | **100.00%** | 97.21% | 89.77% | **100.00%** | 99.23% | 97.42% | 98.97% | **100.00%** | **100.00%** | **100.00%** | **100.00%** |
| 5 | 97.03% | 89.22% | 94.42% | **99.61%** | 87.86% | 95.88% | 82.72% | 96.71% | 93.49% | 86.36% | 99.17% |
| 6 | 99.26% | 99.26% | 97.03% | 99.26% | 56.17% | 76.54% | 90.95% | 99.18% | **100.00%** | 95.87% | **100.00%** |
| 7 | **100.00%** | **100.00%** | **100.00%** | **100.00%** | 98.29% | **100.00%** | 99.57% | **100.00%** | **100.00%** | **100.00%** | **100.00%** |
| 8 | 99.51 | 97.04% | 99.51% | **100.00%** | 93.44% | 99.45% | 98.91% | **100.00%** | **100.00%** | **100.00%** | **100.00%** |
| 9 | 97.13% | 94.27% | 92.99% | **100.00%** | **74.73%** | 95.76% | 94.35% | **100.00%** | **100.00%** | 99.65% | **100.00%** |
| 10 | **100.00%** | **100.00%** | **100.00%** | **100.00%** | 89.96% | 98.21% | 98.66% | 99.11% | **100.00%** | 86.55% | **100.00%** |
| 11 | 97.38% | **100.00%** | 99.02% | **100.00%** | 94.55% | 95.64% | 99.64% | **100.00%** | **100.00%** | 97.08% | **100.00%** |
| 12 | 96.13% | 95.58% | 98.90% | **100.00%** | 78.22% | **100.00%** | 99.39% | **100.00%** | **100.00%** | **100.00%** | 98.77% |
| 13 | 98.13% | 98.51% | 98.13% | 98.53% | 84.50% | **100.00%** | 99.59% | **100.00%** | **100.00%** | **100.00%** | **100.00%** |
| 14 | **93.68%** | 86.32% | 85.26% | **100.00%** | 98.84% | 83.72% | 91.86% | 90.70% | **100.00%** | 95.29% | **100.00%** |
| OA | 98.43±1.65% | 96.83±1.78% | 96.64±1.21% | 99.72±0.89% | 87.10±1.77% | 96.01±2.61% | 96.59±2.47% | 99.32±1.64% | 99.46±1.36% | 97.05±1.48% | **99.66±0.22%** |
| AA | 98.31±1.21% | 96.19±1.32% | 96.26±0.81% | 99.68±0.61% | 87.03±1.34% | 95.84±1.89% | 96.69±2.01% | 98.98±0.91% | 99.53±0.82% | 97.20±0.99% | **99.67±0.19%** |
| Kappa | 98.30±2.01% | 96.56±2.19% | 96.36±1.55% | 99.70±1.03% | 86.01±1.79% | 95.67±2.85% | 96.30±2.94% | 99.26±1.82% | 99.41±1.46% | 96.81±1.70% | **99.63±0.24%** |

* The **best** result is in bold, the second best is underlined. SPEC-F is the abbreviation of Spectralformer. W/O represents without pretrained.

achieving 99.91% OA, 99.92% AA, and 99.91% Kappa, with minimal standard deviations of 0.08%, 0.07%, and 0.09%, respectively.

For the IN dataset, when trained with only 2% of the samples, it achieves OA, AA, and Kappa scores of 81.32%, 86.25%, and 78.97%, respectively, with relatively

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TABLE VII
CLASSIFICATION RESULTS OF DIFFERENT METHODS USING 10% TRAINING SAMPLES PER CLASS ON IP DATASET

| Classes | CVT [62] | Wave-Vit [63] | RVT [64] | CLOLN [65] | HSIC-FM [66] | CSIL [67] | SPEC-F [45] | SS-MTr [68] | HyperSigma [33] | W/O HyperSL | HyperSL |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 84.78% | 93.48% | 50.00% | 88.00% | 82.69% | **100.00%** | **70.59%** | 57.14% | 94.44% | **100.00%** | 95.65% |
| 2 | 95.59% | 93.35% | 76.54% | 89.16% | 63.19% | 89.31% | 81.12% | 95.49% | 97.46% | 92.61% | **98.52%** |
| 3 | 89.28% | 91.57% | 78.07% | 91.27% | 67.30% | 89.73% | 88.59% | 95.85% | 98.19% | 94.65% | **99.46%** |
| 4 | 98.73% | 86.08% | 85.23% | **100.00%** | 89.46% | 99.32% | 95.75% | 90.19% | 99.47% | 97.18% | 98.59% |
| 5 | 94.00% | 88.41% | 52.59% | 98.10% | 90.84% | 95.67% | 92.25% | 98.85% | 96.62% | 95.85% | **99.31%** |
| 6 | 97.53% | **100.00%** | 90.96% | **100.00%** | 93.67% | 99.06% | 96.36% | 99.70% | 98.63% | 99.85% | **100.00%** |
| 7 | **100.00%** | **100.00%** | 92.86% | **100.00%** | 83.33% | 94.44% | 89.00% | 88.46% | **100.00%** | **100.00%** | **100.00%** |
| 8 | **100.00** | **100.00%** | 96.86% | 98.94% | 98.20% | **100.00%** | 64.27% | **100.00%** | **100.00%** | 98.84% | 97.67% |
| 9 | **100.00%** | **100.00%** | **100.00%** | **100.00%** | 80.00% | **100.00%** | 74.11% | 61.11% | 81.25% | **100.00%** | **100.00%** |
| 10 | 96.09% | 93.52% | 87.35% | 95.07% | 77.55% | 93.08% | 98.77% | 92.69% | 96.01% | 91.53% | **96.80%** |
| 11 | 98.17% | 97.47% | 83.95% | 67.59% | 58.82% | 84.57% | 90.11% | 98.01% | **98.73%** | 97.24% | 98.42% |
| 12 | 93.76% | 84.82% | 73.36% | 97.08% | 65.41% | 90.46% | 76.97% | 95.88% | **99.37%** | 93.62% | 96.62% |
| 13 | 95.61% | **100.00%** | 96.10% | **100.00%** | 99.13% | **100.00%** | **100.00%** | **100.00%** | **100.00%** | **100.00%** | **100.00%** |
| 14 | **99.60%** | 97.71% | 95.89% | 98.59% | 90.30% | 95.66% | 82.05% | 98.24% | 99.11% | 98.86% | 98.15% |
| 15 | 94.56% | 96.63% | 82.90% | 97.65% | 65.32% | 95.95% | **100.00%** | 93.10% | 99.03% | 69.16% | 96.54% |
| 16 | 83.87% | 89.25% | 91.40% | **100.00%** | **100.00%** | 98.11% | **100.00%** | 98.81% | **100.00%** | 98.80% | **100.00%** |
| OA | 96.30±1.95% | 94.89±1.89% | 83.45±1.61% | 85.68±1.75% | 73.48±2.95% | 91.43±2.22% | 79.49±3.68% | 96.53±1.11% | 98.30±0.87% | 95.04±1.78% | **98.48±0.39%** |
| AA | 95.10±1.64% | 94.52±1.21% | 83.38±1.26% | 95.09±1.23% | 81.57±2.57% | 95.43±1.53% | 87.50±2.77% | 91.47±1.04% | 97.39±0.53% | 95.51±1.32% | **98.70±0.25%** |
| Kappa | 95.78±2.47% | 94.16±2.03% | 81.15±1.88% | 83.31±1.79% | 69.92±3.46% | 90.18±2.86% | 76.81±3.94% | 96.05±1.17% | **98.30±0.91%** | 94.34±1.86% | 98.27±0.45% |

* The **best** result is in bold, the second best is underlined. SPEC-F is the abbreviation of Spectralformer. W/O represents without pretrained.

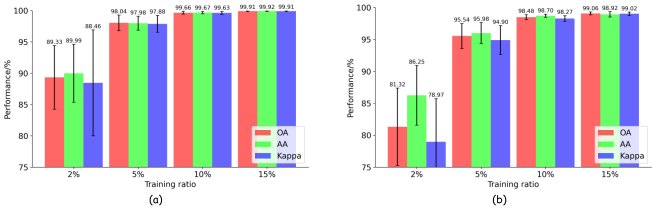



Fig. 16. Classification results under different training sample proportions. (a) BOS. (b) IP.

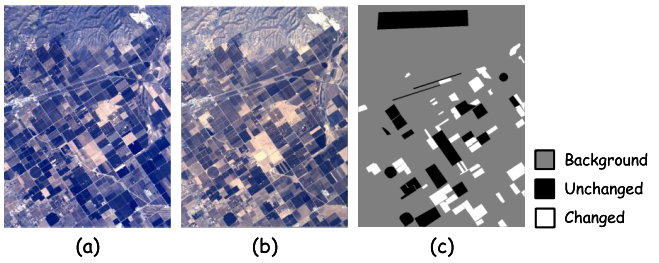



Fig. 17. Pseudocolor images and groundtruth of the SantaBarbara dataset. (a) SantaBarbara in 2013. (b) SantaBarbara in 2014. (c) Groundtruth.

high standard deviations of 6.06%, 4.67%, and 6.76%, indicating variability in performance. With 5% of the training samples, the model shows a notable increase in accuracy, achieving 95.54% OA, 95.98% AA, and 94.90% Kappa, while the standard deviations decrease to 1.96%, 1.65%, and 2.26%, reflecting enhanced stability. As the sample size expands to 10%, the model further improves, reaching 98.48% OA, 98.70% AA, and 98.27% Kappa, with minimal standard deviations of 0.39%, 0.25%, and 0.45%, suggesting more consistent performance. When the training ratio reaches 15%, it can be observed that the model has stabilized, where the model achieves 99.06% OA, 98.92% AA, and 99.02% Kappa, with standard deviations reduced to 0.23%, 0.42%, and 0.26%, respectively. These results indicate that the model maintains high classification accuracy even under limited sample conditions. As the number of training samples increases, both accuracy and stability improve progressively. Notably, when the training sample proportion reaches 10%, the model achieves the optimal trade-off between performance and efficiency.

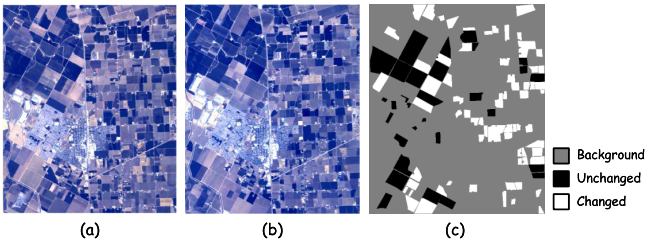



Fig. 18. Pseudocolor images and groundtruth of the BayArea dataset. (a) BayArea in 2013. (b) BayArea in 2015. (c) Groundtruth.

### C. Change Detection

Change detection is another fundamental task in hyperspectral analysis. In this section, we fine-tune the change detection model using the pretrained model and conduct ablation analysis, similar to the classification task, to assess the effectiveness of pretrained weights and the impact of spectral position encoding on detection accuracy.

*1) Dataset Description:* We used two commonly used benchmark datasets for testing: Barbara, and BayArea. The Pseudocolor and groundtruth of them are shown in Figs. 17 and 18, respectively. The Barbara scene, captured in 2013 and 2014 by the AVIRIS sensor over the Barbara region (California), has spatial dimensions of 984 × 740 pixels and includes 224 spectral bands. The BayArea scene, captured in 2013 and 2015 by the AVIRIS sensor around the city of Patterson (California), has spatial dimensions of 600 × 500 pixels and includes 224 spectral bands. The central wavelength information for Barbara and BayArea was obtained from the AVIRIS Data Products Portal. The Barbara dataset comprises 52 134 changed pixels and 80 418 unchanged pixels, while the BayArea dataset consists of 38 425 changed pixels and 34 211 unchanged pixels. We randomly select 1% of the dataset as the training set, with the remaining samples designated as the test set.

*2) Implement Details:* Similar to the classification network, in the change detection task, we only use the encoder part of HyperSL to generate a unified representation of the input hyperspectral data. First, patches from two temporal phases are flattened and input into HyperSL

$$H_{f1} = \text{HyperSL_encoder}(X_{f1}) \tag{19}$$

$$H_{f2} = \text{HyperSL_encoder}(X_{f2}). \tag{20}$$

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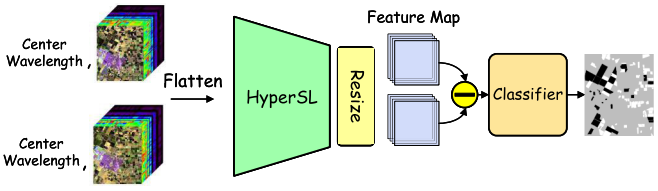



Fig. 19. Overall architecture of HyperSL for change detection task.

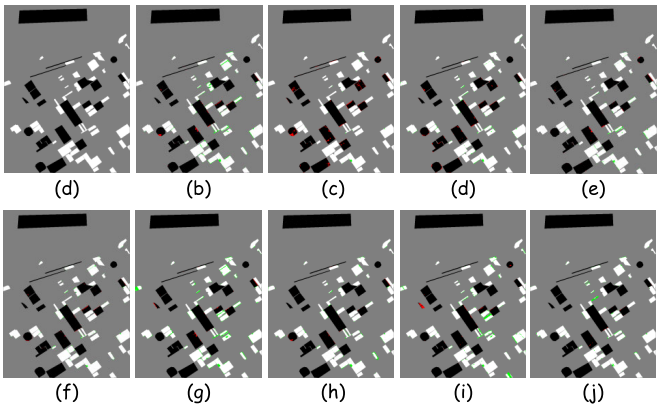



Fig. 20. Change detection results of different methods on the SantaBarbara dataset. (a) Groundtruth. (b) CSANet. (c) DIEFEN. (d) GTMSiam. (e) SSTFormer. (f) MHCD. (g) MSCSCNet. (h) HyperSigma. (i) W/O HyperSL. (j) HyperSL. W/O represents without pretrained.

Next, the output features are resized into feature maps, where $H_{f1} \rightarrow H_1, H_1 \in R^{p\times p\times d}$ and $H_{f2} \rightarrow H_2 \in R^{p\times p\times d}$. Then, the absolute difference between the features of the two temporal phases is taken as the input for the subsequent change detection network

$$H = |H_1 - H_2|. \tag{21}$$

The change detection network consists of a simple convolutional classifier, with operations as follows:

$$H' = \text{GAP}(\text{ConvBlock}(H)) \tag{22}$$

$$\text{logits} = \text{MLP}(H'). \tag{23}$$

Finally, binary cross-entropy loss is used for end-to-end fine-tuning optimization of the model

$$\theta^* = \arg\min_{\theta} \text{Binary Cross Entropy}(\text{logits}, y). \tag{24}$$

The overall architecture of change detection network is illustrated in Fig. 19. In our experiments, we tested the Barbara dataset, the Hermiston city dataset, and another Hermiston city dataset, setting the patch size to 5 for each. Ten percent of each class was used as training samples, with the remaining samples as the test set. The model was trained for 100 epochs, with the learning rate set to 0.0001, and the AdamW optimizer was used with a weight decay of 0.05. We evaluated change detection performance using OA, Kappa, $F1$, Precision, and Recall as metrics. We compared our approach with current mainstream and advanced change detection methods, which include CSANet [69], DIEFEN [70], GTMSiam [71], SSTFormer [72], MHCD [73], MSCSCNet [74], and HyperSigma [33].

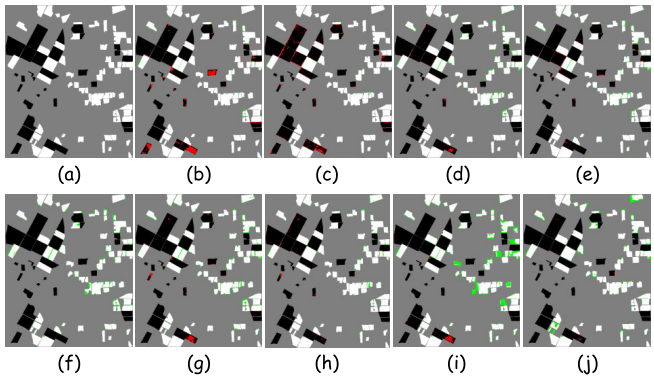



Fig. 21. Change detection results of different methods on the BayArea dataset. (a) Groundtruth. (b) CSANet. (c) DIEFEN. (d) GTMSiam. (e) SSTFormer. (f) MHCD. (g) MSCSCNet. (h) HyperSigma. (i) W/O HyperSL. (j) HyperSL. W/O represents without pretrained.

TABLE VIII
CHANGE DETECTION RESULTS OF VARIOUS METHODS ON THE BARBARA DATASET WITH 1% TRAINING SAMPLES

| Methods | OA | Kappa | F1 | Pr | Re |
|---|---|---|---|---|---|
| CSANet [69] | 97.16% | 94.01% | 98.35% | 94.36% | 96.32% |
| DIEFEN [70] | 97.78% | 94.51% | 97.26% | 95.19% | 99.41% |
| GTMSiam [71] | 97.92% | 95.63% | 97.35% | 97.49% | 97.21% |
| SSTFormer [72] | 98.41% | 96.66% | 97.96% | 97.23% | 97.96% |
| MHCD [73] | **98.72%** | 97.32% | **98.95%** | 99.30% | 98.61% |
| MSCSCNet [74] | 96.12% | 93.71% | 94.88% | **98.66%** | 91.38% |
| HyperSigma [33] | 98.58% | 98.34% | 97.61% | 97.92% | 97.73% |
| W/O HyperSL | 96.54% | 95.94% | 95.47% | 92.80% | 98.30% |
| HyperSL | 98.66% | **98.42%** | 98.28% | 96.97% | **99.62%** |

* The **best** result is in bold, the second best is underlined. W/O represents without pretrained.

TABLE IX
CHANGE DETECTION RESULTS OF VARIOUS METHODS ON THE BAYAREA DATASET WITH 1% TRAINING SAMPLES

| Methods | OA | Kappa | F1 | Pr | Re |
|---|---|---|---|---|---|
| CSANet [69] | 95.05% | 90.01% | 93.03% | 94.36% | 98.09% |
| DIEFEN [70] | 96.16% | 92.25% | 96.51% | 93.88% | **99.29%** |
| GTMSiam [71] | 96.45% | 92.88% | 96.62% | 98.30% | 95.00% |
| SSTFormer [72] | **96.97%** | 93.93% | **97.14%** | 96.10% | 98.20% |
| MHCD [73] | 96.07% | 92.15% | 95.95% | **99.85%** | 92.34% |
| MSCSCNet [74] | 95.25% | 91.03% | 95.47% | 97.30% | 93.71% |
| HyperSigma [33] | 96.31% | 92.07% | 96.64% | 99.71% | 92.12% |
| W/O HyperSL | 90.84% | 88.67% | 90.80% | 84.61% | 97.96% |
| HyperSL | 96.81% | **95.56%** | 96.31% | 97.71% | 97.00% |

* The **best** result is in bold, the second best is underlined. W/O represents without pretrained.

*3) Results and Ablation Analysis:* Tables VIII and IX present the change detection results of different methods on the Barbara and BayArea datasets, respectively. The result visualizations are shown in Figs. 20 and 21, respectively.

For the Barbara dataset, our method achieves OA, Kappa, $F1$, Precision, and Recall scores of 98.66%, 98.42%, 98.28%, 96.97%, and 99.62% respectively. Compared to other methods, our method demonstrates superior performance, achieving the highest scores in Kappa and Recall, while ranking second in OA. For the BayArea dataset, our method attains OA, Kappa, $F1$, Precision, and Recall scores of 96.81%, 95.56%, 96.31%, 97.71%, and 97.00%, respectively. It achieves the best performance in Kappa, while ranking second in OA

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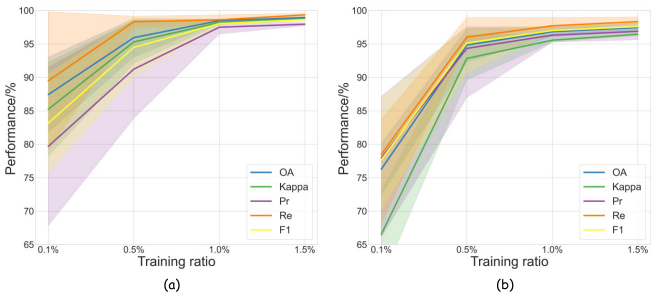



Fig. 22. Change detection results under different training sample proportions. (a) Barbara. (b) BayArea.

among the evaluated methods. These results demonstrate the robustness and effectiveness of our method in cross source HSI change detection task. Notably, we achieve performance comparable to other mainstream methods across change detection tasks for different data source by applying only simple convolution operations to the dual-temporal features output by the HyperSL encoder. The underlying reason for this effectiveness lies in the encoder's ability to extract robust, domain-invariant spectral representations, which enhances its generalization across different data sources. Specifically, the spectral position encoding mechanism enables the model to better align distributional discrepancies between multisource HSIs, ensuring more stable feature learning. Furthermore, the extracted feature representations emphasize intrinsic structural and temporal changes rather than domain-specific variations, allowing the convolutional operations to effectively capture meaningful changes while reducing sensitivity to noise and inconsistencies between datasets. This ensures that the model remains adaptable and reliable in diverse hyperspectral change detection scenarios.

In addition, we perform ablation experiments using randomly initialized network weights to analyze the impact of pretraining on improving downstream change detection tasks. The randomly initialized HyperSL achieves OA, Kappa, $F1$, Precision, and Recall scores of 96.54%, 95.94%, 95.47%, 92.80%, and 98.30% on the BOS dataset respectively, and 90.84%, 88.67%, 90.80%, 84.61%, and 97.96% on the BayArea dataset, respectively. By initializing the model with pretrained weights, the performance on the Barbara dataset improves by 2.12%, 2.93%, 2.81%, 4.17%, and 1.32% for OA, Kappa, $F1$, Precision, and Recall, respectively. On the BayArea dataset, the improvements are 5.97%, 6.89%, 5.51%, 13.1%, and 0.96% for the same metrics. These improvements strongly demonstrate the effectiveness of our pretrained weights in enhancing performance for change detection tasks.

Moreover, we conducted ten independent experiments on Barbara and BayArea datasets with different proportions of training samples to examine the stability and robustness of the model in the change detection task, as illustrated in Fig. 22. For the Barbara dataset, with 0.1% training data, the OA, Kappa, $F1$, Precision, and Recall scores are 87.43%, 85.25%, 79.68%, 89.49%, and 83.17%, respectively, with standard deviations of 5.61%, 7.01%, 11.83%, 10.34%, and 7.41%. When the training proportion increases to 0.5%, the scores improve substantially to 95.94%, 95.30%, 91.23%, 98.32%, and 95.47%, while the corresponding standard deviations decrease to 2.87%, 3.20%, 7.34%, 0.85%, and 4.35%, suggesting enhanced robustness under low-sample conditions. With 1% training data, the model achieves further improvement, attaining 98.66% OA, 98.42% Kappa, 98.28% $F1$, 96.97% Precision, and 99.62% Recall, with reduced standard deviations of 0.23%, 0.27%, 1.00%, 0.76%, and 0.30%, demonstrating that the model has reached a stable performance state. At 1.5% training proportion, the model achieves near-optimal performance with 98.94% OA, 98.74% Kappa, 97.95% $F1$, 99.34% Precision, and 98.64% Recall, while standard deviations further decrease to 0.18%, 0.21%, 0.22%, 0.24%, and 0.23%, indicating both high accuracy and stability.

For the BayArea dataset, with only 0.1% training data, the OA, Kappa, $F1$, Precision, and Recall scores are 76.28%, 66.55%, 77.97%, 78.49%, and 77.66%, respectively, with standard deviations of 3.83%, 6.55%, 11.83%, 10.34%, and 7.41%, indicating notable performance fluctuations under extremely limited data conditions. At 0.5% training proportion, the model demonstrates a substantial improvement, achieving 94.84% OA, 92.83% Kappa, 94.33% $F1$, 96.02% Precision, and 95.13% Recall, with reduced standard deviations of 2.51%, 3.50%, 3.25%, 2.92%, and 2.36%, suggesting enhanced stability. With 1% training data, the model further refines its accuracy, reaching 96.81% OA, 95.56% Kappa, 96.31% $F1$, 97.71% Precision, and 97.00% Recall, while the corresponding standard deviations decrease to 0.67%, 0.95%, 1.27%, 1.34%, and 0.64%, indicating improved robustness. At 1.5% training proportion, the model attains near-optimal performance, with 97.45% OA, 96.45% Kappa, 96.88% $F1$, 98.33% Precision, and 97.60% Recall, while standard deviations further decrease to 0.63%, 0.87%, 1.25%, 0.72%, and 0.61%, highlighting the model's strong accuracy and consistency. Overall, while the model exhibits performance fluctuations when trained with extremely limited samples, its classification accuracy and stability significantly improve as more training data is introduced. When 1% of the training samples are available, the model already achieves strong performance, with further increases in training data yielding only marginal improvements.

### *D. Unsupervised Classification and Change Detection*

To enhance the comprehensiveness of our evaluation, we evaluate the model's performance through unsupervised change detection and unsupervised classification for large-scale HSI dataset.

*1) Dataset Description:* For unsupervised change detection, we continue to use Barbara and BayArea as benchmark data. For the unsupervised classification, we use the Chikusei dataset, which was acquired using the Headwall Hyperspec-VNIR-C sensor over Chikusei City, Japan. The dataset consists of 128 spectral bands covering the wavelength range of 343–1018 nm, with a high spatial resolution of 2.5 m/pixel. The HSI has a size of 2517 × 2335 pixels and includes a diverse set of 19 classes, encompassing both urban and rural environments.

*2) Implement Details:* For the unsupervised change detection, the overall model structure is similar to that of supervised

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TABLE X
UNSUPERVISED CHANGE DETECTION RESULTS OF BARBARA DATASET

| Methods | OA | Kappa | F1 | Pr | Re |
|---|---|---|---|---|---|
| CVA [75] | 83.26% | 65.24% | 79.26% | 77.30% | 81.32% |
| DCVA [76] | 84.33% | 66.88% | 79.57% | 81.68% | 77.57% |
| MSCD [77] | 78.68% | 53.13% | 68.72% | 81.27% | 59.52% |
| OSCD [78] | 68.69% | 33.84% | 59.14% | 61.37% | 57.51% |
| MTC-NET [79] | 88.05% | 75.50% | 85.66% | 82.88% | 89.20% |
| HyperNet [80] | 91.14% | 81.48% | 88.80% | 88.36% | 89.25% |
| HyperSL | **91.44%** | **82.07%** | **89.15%** | **89.44%** | **88.85%** |

* The **best** result is in bold, the second best is underlined.

TABLE XI
UNSUPERVISED CHANGE DETECTION RESULTS OF BAYAREA DATASET

| Methods | OA | Kappa | F1 | Pr | Re |
|---|---|---|---|---|---|
| CVA [75] | 84.68% | 69.61% | 84.27% | 93.35% | 76.80% |
| DCVA [76] | 84.53% | 69.25% | 84.36% | 91.77% | 78.05% |
| MSCD [77] | 75.03% | 92.25% | 72.60% | 87.77% | 61.90% |
| OSCD [78] | 72.64% | 50.89% | 73.39% | 86.44% | 70.65% |
| MTC-NET [79] | 88.21% | 0.7627% | 89.10% | 88.07% | 90.16% |
| HyperNet [80] | **90.79%** | **81.52%** | **91.29%** | **92.24%** | **90.37%** |
| HyperSL | 84.97% | 70.33% | 83.99% | 73.75% | 97.52% |

* The **best** result is in bold, the second best is underlined.

change detection, with the only modification being the replacement of the final classification layer with a clustering layer, where the parameters of the HyperSL are frozen. We evaluated unsupervised change detection performance using OA, Kappa, $F1$, Precision, and Recall as metrics and compared our approach with current advanced change detection methods, including: CVA [75], DCVA [76], MSCD [77], OSCD [78], MTC-NET [79], and HyperNet [80].

For the unsupervised classification task, we consider only spectral signals and apply clustering to the spectral representations extracted by HyperSL, where the parameters are frozen. To evaluate the capability of HyperSL in unsupervised classification, we conduct comparative experiments using different feature representations, including raw data, PCA-reduced data, autoencoder-extracted features, and HyperSL-derived representations. Each of these feature sets is clustered using $K$-means, spectral clustering (SC), and agglomerative clustering (AC), enabling a comprehensive comparison of clustering performance across different representations. The performance unsupervised classification is evaluated by OA, AA, Kappa, adjusted rand index (ARI), and normalized mutual information (NMI).

*3) Results and Ablation Analysis:* Tables X and XI present the unsupervised change detection results of different methods on the Barbara and BayArea datasets, respectively.

On the Barbara dataset, our method achieves OA, Kappa, $F1$, Precision, and Recall scores of 91.44%, 82.07%, 89.15%, 89.44%, and 88.85% respectively. Compared to other methods, our method demonstrates superior performance, achieving the highest scores in all metrics. This superior performance can be attributed to the large scale self-supervised pretraining, which enables the extraction of robust spectral representations, even when the network is frozen during clustering.

TABLE XII
UNSUPERVISED CLASSIFICATION RESULT OF CHIKUSEI DATASET

| Methods | OA | AA | Kappa | ARI | NMI |
|---|---|---|---|---|---|
| Kmeans | 42.69% | 36.02% | 37.50% | 0.2890 | 0.5824 |
| PCA+Kmeans | 41.11% | 45.21% | 35.93% | 0.2757 | 0.5812 |
| AE+Kmeans | 39.44% | 37.38% | 33.05% | 0.2798 | 0.5663 |
| HyperSL+Kmeans | 60.23% | 52.04% | 56.16% | 0.5405 | 0.7425 |
| SC | 61.56% | 58.99% | 55.72% | 0.4568 | 0.7244 |
| PCA+SC | 52.97% | 50.60% | 46.45% | 0.3452 | 0.6212 |
| AE+SC | 50.13% | 41.88% | 43.44% | 0.3504 | 0.6260 |
| HyperSL+SC | 70.93% | 59.15% | 66.67% | 0.6565 | 0.7854 |
| AC | 47.01% | 45.06% | 41.44% | 0.3477 | 0.6325 |
| PCA+AC | 44.55% | 37.58% | 38.86% | 0.3181 | 0.6179 |
| AE+AC | 41.36% | 41.70% | 35.81% | 0.2883 | 0.5963 |
| HyperSL+AC | **73.43%** | **61.70%** | **70.04%** | **0.6756** | **0.8303** |

*The **best** result is in bold, the second best is underlined. SC represents the Spectral Clustering and AC represents Agglomerative Clustering.

By leveraging pretraining on large-scale hyperspectral data, HyperSL captures meaningful spectral structures and generalizes well to unsupervised clustering tasks. However, on the BayArea dataset, our method attains OA, Kappa, $F1$, Precision, and Recall scores of 84.97%, 70.33%, 83.99%, 73.75%, and 97.52%, respectively. Compared to other methods, our method ranks third across all evaluation metrics. This decline in performance may be attributed to the higher complexity of land-cover types in the BayArea, which includes urban, coastal, and vegetation regions, leading to more ambiguous spectral boundaries. Additionally, the spectral characteristics of the BayArea dataset may differ from those in the pretraining corpus, affecting the generalization of HyperSL's learned features. Unlike end-to-end trained deep learning methods such as MTC-NET and HyperNet, which may have been trained on this specific dataset, HyperSL relies entirely on pretrained features without additional adaptation, potentially limiting its effectiveness when the test data distribution deviates from the pretraining set.

Table XII presents the experimental results of different feature representation methods combined with various clustering algorithms, and it clearly demonstrates the superiority of HyperSL over traditional feature extraction methods, particularly PCA and AE, in the context of unsupervised classification.

Across all clustering algorithms, HyperSL consistently achieves the highest performance, with HyperSL + ac attaining an OA of 73.43% and HyperSL + SC achieving 70.93%, significantly outperforming other feature representations. Even with $K$-means, which typically struggles with nonlinear separability, HyperSL improves the unsupervised classification performance to 60.32% OA, demonstrating its robustness. This outstanding performance can be attributed to HyperSL's pretraining on large-scale spectral data, which allows the model to learn rich spectral representations that are invariant to noise and robust across different data distributions. Unlike PCA and AE, which are only trained or fit on the experimental dataset itself, HyperSL benefits from a large-scale self-supervised learning pretraining phase, enabling it to capture meaningful spectral relationships beyond the limited

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training set. The fact that HyperSL achieves superior performance while keeping the network frozen during unsupervised classification further highlights the generalization capability of its learned spectral representations. In contrast, PCA and AE exhibit a noticeable decline in the unsupervised classification performance, mainly because their transformations are limited to the experimental dataset and do not generalize well. PCA, as a linear transformation method, projects data onto principal components based on variance maximization but lacks the ability to capture complex spectral structures inherent in HSIs. Since it is only computed from the given dataset, PCA cannot leverage prior spectral knowledge, leading to degraded unsupervised classification performance. Similarly, AE-extracted features also suffer from performance degradation because the AE is trained purely on the experimental dataset using a reconstruction loss, rather than leveraging prior spectral knowledge like HyperSL. Since AE is optimized to reconstruct the input data, it tends to focus on minimizing pixel-wise errors rather than learning discriminative spectral embeddings. This explains why AE-extracted features perform poorly, as the extracted features lack global spectral structure and tend to overfit to local patterns in the dataset.

Overall, these results highlight the fundamental limitation of dataset-dependent feature extraction methods like PCA and AE. Since HyperSL leverages large-scale spectral pretraining, it learns more generalizable and discriminative representations, making it significantly more effective for HSI unsupervised classification compared to feature extraction techniques trained solely on the experimental dataset.

## V. Conclusion

Hyperspectral signals can be regarded as a form of "natural language" through which materials express their intrinsic properties. Building upon this concept, we propose HyperSL, a novel foundational model for HSIs. It transforms any hyperspectral signals into token sequences analogous to word embeddings, enabling unified processing through a Transformer-based architecture. This universality means that, for HSI interpretation tasks, our model can leverage information from other datasets, allowing for greater flexibility and effectiveness in different downstream tasks. To preserve critical wavelength information lost during digitization, we introduce spectral wavelength encoding for the first time, ensuring the retention of physical wavelength characteristics. HyperSL is pretrained on a large scale, leveraging a masked reconstruction self-supervised learning paradigm over 300 million spectral instances collected from EnMAP, DESIS, and a private airborne hyperspectral dataset, covering diverse spectral ranges and imaging conditions. The unique architecture of HyperSL allows it to handle data with varying spectral band numbers, while its spectral wavelength encoding enables self-alignment of spectral features across different spectral ranges during training. This equips HyperSL with strong cross-domain generalization, making it highly adaptable for processing hyperspectral datasets acquired from different sensors. A key advantage of HyperSL lies in its ability to transfer knowledge seamlessly to diverse downstream tasks without requiring architectural modifications or retraining from scratch. We rigorously evaluate its effectiveness on both supervised and unsupervised learning paradigms, encompassing classification and change detection across multiple benchmark datasets. For supervised classification, using only 10% of labeled samples, HyperSL achieves 99.66% OA on the BOS dataset and 98.48% OA on the IP dataset. In unsupervised classification, hierarchical clustering on HyperSL-extracted representations attains 73.43% OA on the Chikusei dataset. For change detection, HyperSL demonstrates state-of-the-art performance, achieving 98.66% and 96.81% OA on the Barbara and BayArea datasets, respectively, when fine-tuned with just 1% labeled data. Furthermore, in fully unsupervised inference, it achieves 91.44% OA on Barbara and 84.97% OA on BayArea. These results demonstrate that HyperSL consistently achieves outstanding performance across multiple datasets and tasks, highlighting its exceptional generalization and transferability. Moving forward, we aim to further explore its applicability across a broader range of hyperspectral tasks, refine its architectural efficiency, and optimize its tokenization strategy to reduce computational complexity while maintaining state-of-the-art performance.

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